Ahmed Alaa

Unity Game Developer

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Gharbia, Egypt · Open to relocation (UAE, KSA)

SUMMARY

Unity Game Developer with 1+ year of professional experience and 2 years teaching Unity and C#. Shipped a Meta Quest 3 VR title and a bilingual (EN/AR) multiplayer training product on Photon PUN 2. Builds decoupled, data-driven gameplay and simulation systems with ScriptableObjects, state machines, interfaces and object pooling. Hosted the game development session at NASA Space Apps Shebeen El-Kom. Experienced with Git, Agile/Scrum, and AI-assisted development including Claude Code.

TECHNICAL SKILLS

Unity Engine — Unity 6, URP, Cinemachine, Addressables, Input System, Localization, Android & WebGL builds
Programming — C# (OOP), Lua, JavaScript (three.js), HTML & CSS
Architecture — Strategy, State & Object Pool patterns, Interface Contracts, ScriptableObject Architecture, Assembly Definitions, SOLID
Performance — Object pooling, allocation-free physics queries, event-driven systems over per-frame polling
Simulation & Math — Ballistics & Fire Control, Lead Targeting & Interception
Multiplayer & Data — Photon PUN 2, Firebase Realtime Database, JSON Data Management, RTL Localization (Arabic/English)
Tools — Git, GitHub, GitLab, Fork, Blender, Substance Painter, Premiere Pro, Photoshop, Roblox Studio (6 mo.)

PROFESSIONAL EXPERIENCE

Junior Unity Game Developer, Zinad 07/2025 – Present | Remote

- Developed a VR rhythm-action game for Meta Quest 3 — migrated from Oculus SDK 0.46.0 to Meta XR All-in-One SDK 72.0.0, rebuilt the player prefab using Building Blocks, and resolved Android build failures caused by a TMP3D shader incompatibility with GLES3 multiview stereo rendering
- Implemented dual-saber hit detection with a same-frame double-hit guard via Script Execution Order, and a DSP clock-based spawner syncing object pool releases to BPM
- Built a ScriptableObject-based state machine handling gameplay/quiz transitions with fully decoupled cross-system communication, and a quiz system with DOTween crossfade transitions and live Arabic/English locale switching
- Shipped ZiRunner, a bilingual (EN/AR) multiplayer endless runner for security-awareness training — outcome-authoritative Photon PUN 2 netcode, RTL Arabic localization, and a StreamingAssets JSON content pipeline editable post-ship without a rebuild

Game Development Instructor, Da3em Academy 05/2023 – 08/2025 | Remote

- Designed and taught a full Unity and Roblox curriculum, mentoring students from first concept to their first launched games

PROJECTS

MilSim — Military Simulation Framework
Modular Unity 6 framework of military systems (C-RAM air defence, drone swarms, 2A65 Msta-B howitzer) that interact only through shared contracts, with assembly-enforced boundaries. Plugged a howitzer's shells into the air defence without editing it, then measured, diagnosed and fixed crossing-shot leaks: raised interception from 76% / 67% to 100%.

Emplacement · github.com/Ahmed-A-Abdou/emplacement
Open-source Unity package (MIT, UPM): StateMachine<T>, ObjectPool<T>, cross-prefab references. Extracted from MilSim; reused unchanged in a second project.

Tunable Solar System
Playable WebGL demo for my NASA Space Apps session: Kepler orbits as swappable strategy assets, data-driven bodies, built on Emplacement.

Cam3
Solo horror game. 7-state enemy FSM over a weighted node graph, event-driven ScriptableObject architecture.

SPEAKING & LEADERSHIP

Host, Game Development Session — NASA Space Apps Challenge, Shebeen El-Kom 10/2026

- Delivered “Build Systems, Not Spaghetti”, a 50-minute session for beginners on data-driven architecture, state machines and the strategy pattern, with live Unity demos

CIS Team MU — Team Leader & Game Development Instructor 09/2023 – 09/2024

- Led one of the university's largest student organizations (700+ members), running 10+ large events and workshops
- Trained 50+ members in Unity and C# through game jams (15+ prototypes); spoke at Beyond Earth, MagiCoders and VICTORIS 4.0

EDUCATION

Bachelor's Degree in Software Engineering 09/2021 – 07/2025
Faculty of Computer and Information Sciences, Mansoura University

AWARDS

1st Place, Advanced Level — Game Development Challenge, ITI TechAdventure (ITI, Cairo University) 01/2025

LANGUAGES

Arabic (Native) · English (Fluent – C2)